VIETNAM GENERAL CONFEDERATION OF LABOR
TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY



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FINAL REPORT

SOFTWARE ENGINEERING

SAIGONBITE
Cloud-Kitchen Food Delivery Hub for Ho Chi Minh City

Part 1: Project Proposal and Requirements Engineering

Technology Tier: Tier 4 (maximum 10.0)

HO CHI MINH CITY, 2026

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Advised by
THS. PHẠM THÁI KỲ TRUNG

HO CHI MINH CITY, 2026

ACKNOWLEDGEMENT

We thank our lecturer, ThS. Phạm Thái Kỳ Trung, for the SaigonBite case study, project specification and student survival guide. These materials define the business problem and the engineering decisions that this proposal must address.

This report sets out the proposed scope, individual use case ownership, measurable requirements and working assumptions for Part 1. We acknowledge the course sources in REFERENCES and disclose assistance from generative AI in Appendix A.

Ho Chi Minh City, 25/10/2026
Authors
(Signature and full name)

Ngô Lâm Tiến Nguyễn Công Thành Nguyễn Trọng Phú

DECLARATION OF AUTHORSHIP

We submit this Part 1 proposal for Software Engineering, Sem01_2627, Ton Duc Thang University. The business rules BR1–BR18 originate from the course specification; our refined requirements and working assumptions are identified separately.

Generative AI assistance and the availability of supporting records are disclosed in Appendix A. The traceability matrix describes planned design and verification work; it does not certify that the application or its tests have passed.

The team is responsible for checking the submitted content, explaining its requirements and decisions, and declaring all assistance. No signatures, instructor approvals or execution results are supplied on behalf of the members.

Ho Chi Minh City, 25/10/2026
Authors
(Signature and full name)

Ngô Lâm Tiến Nguyễn Công Thành Nguyễn Trọng Phú

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CHAPTER 1. INTRODUCTION

1.1 Problem Statement

SaigonBite is a planned network of delivery-only kitchens in Ho Chi Minh City. Each hub operates several virtual food brands. Customers need one place to browse a hub's menu, see a transparent price, pay and track an order; staff need a controlled process from placement to delivery.

The client faces uneven kitchen demand and two customer groups with different identity and payment needs. Quiet kitchens should attract demand through a discount, while hubs at or above 90% load must refuse new orders. Pricing, payment eligibility and status transitions must be enforced consistently on the server, not left to interface behaviour.

1.2 Customers and Stakeholders

Local Customers register with a Vietnamese phone number and simulated OTP and use MoMo, VNPay or COD. Foreign Tourists register with passport, nationality, email and a simulated email code and use Apple Pay, PayPal or COD. Kitchen Staff work at one hub; assigned Shippers perform delivery; System Admins manage catalogue, capacity, vouchers, accounts, rain settings and reports. The two customer types share one database schema.

1.3 Scope and Deliverables

Part 1 contains the proposal, requirements baseline, measurable quality attributes, working ambiguity decisions and an initial traceability matrix. It proposes three main CRUD use cases, with secondary scopes covering the remaining system functions. R1 Revenue and R2 Hub Utilization are mandatory. Detailed UML, the Figma prototype, implementation, sandbox evidence and execution results belong to subsequent parts.

Sources: Project Specification sections 1–4 and Case Study sections 1–4. Requirements are proposed obligations, not claims that implementation is complete.

CHAPTER 2. PROJECT PROPOSAL

2.1 SDLC Methodology and Schedule

We propose Scrum with a prioritised backlog, a short daily coordination meeting, reviews at deliverable milestones and retrospectives after each review. The integration lead coordinates priorities; all three members develop and review their owned scopes. This lightweight allocation suits a three-person academic team and allows the Week 5 change request to enter the backlog without obscuring existing requirements.

Period	Goal and milestone
Weeks 1–2	Analyse requirements, assign ownership and agree the proposed tier. Part 1 report, workbook and pitch due 26 Oct 2026, 07:00 UTC+7.
Weeks 3–4	Model assigned use cases, architecture, state machine and Figma UI. Part 2 due 9 Nov 2026, 07:00 UTC+7.
Weeks 5–6	Show a running skeleton at the Week 5 checkpoint; analyse the issued change request; implement and test the use cases.
Week 7	Complete Part 3 code/QA, traceability and product demo. Deadline: 30 Nov 2026, 07:00 UTC+7.
Final consolidation	Apply feedback and compile the final report/video. Survival Guide separately lists 7 Dec 2026; confirm the final portal instruction when published.

The analysis-to-Part-3 plan spans seven weeks. The separate final consolidation date is not represented as an additional week inside those seven weeks. The current Part 1 portal deadline takes precedence for this submission.

2.2 Team Roles and Ownership

Member	Main use case, role and secondary scope
Ngô Lâm Tiến 524H0195	D — Place and Manage Orders. C - Manage Vouchers; Pricing Engine (BR1-BR12); Payment Strategy Module. Reports: none; owns shared pricing/payment infrastructure.
Nguyễn Công Thành 524H0191	A — Manage Menu & Brands. E - Manage User Accounts. Reports: R1 - Revenue Report.
Nguyễn Trọng Phú 524H0179	B — Manage Hubs & Capacity. F - Manage Deliveries; Change Request Impact Note. Reports: R2 - Hub Utilization Report.

Ngô Lâm Tiến (524H0195) is the named owner of D and the shared pricing/payment interface. Nguyễn Công Thành (524H0191) owns A and R1. Nguyễn Trọng Phú (524H0179) owns B and R2. Secondary scopes C, E and F remain assigned to these same owners. Each member must claim the main use case in the live pitch; this assignment is planned responsibility, not evidence of completed contribution.

2.3 Complete CRUD Boundaries

Use case	Create, Read, Update and Delete meaning
D — Orders	Create a priced order; read/track owned orders; edit items only in Placed; cancel in Placed or Confirmed. Delete means business cancellation, preserving history.
A — Menu and Brands	Create brands and menu items; browse/filter; edit details and availability; soft-delete brands/items while preserving historical order data.
B — Hubs and Capacity	Create hubs with positive integer capacity; read hubs/current load; edit address and capacity; soft-delete hubs while preserving order/report history.

2.4 Proposed Tier and Architecture

The instructor technology-tier table supplied for this revision gives the ceilings below. These are maximum scores, not awarded marks. Source: instructor table archived at docs/references/instructor-technology-tiers.png; supplied by the student on 24 Oct 2026.

Tier	Max /10	Required stack and architecture
Tier 1	7.0	C# WinForms + local SQL database / Excel. 3-Tier architecture.
Tier 1B	7.5	Core PHP (or CodeIgniter) + MySQL + Bootstrap. MVC architecture.
Tier 2	8.5	ASP.NET Web Forms/MVC (Database First) + SQL Server + Bootstrap. MVC.
Tier 3	9.5	ASP.NET MVC (Code First) + Entity Framework + Bootstrap. MVC.
Tier 4	10.0	Tier 3 plus live sandbox API, interactive Admin dashboard and automated tests.

The proposed target remains Tier 4, with a 10.0/10 ceiling: the full Tier 3 foundation plus all three Tier 4 additions. We propose ASP.NET Core MVC (.NET 8), EF Core Code-First, SQL Server and Bootstrap 5. The table names ASP.NET MVC and Entity Framework without explicitly naming Core; instructor confirmation of this variant is pending. No approval or score is inferred. A mock payment does not meet the live sandbox criterion; the Tier 3 ceiling of 9.5 applies only when its foundation is accepted and delivered.

Tier criterion	Planned delivery, owner and acceptance evidence
Tier 3 foundation	ASP.NET Core MVC, EF Core Code-First migrations, SQL Server and Bootstrap 5. Retain the MVC layers; confirm the Core variant with the instructor because the table names ASP.NET MVC. **Owner:** 524H0191. **Evidence:** Part 2 architecture/ERD and migration design; Part 3 reproducible SQL Server setup, migration history and complete CRUD demonstration. SQLite is a local fallback, not the submitted database baseline. **Trace:** UC A/B/D; NFR3/4. Status: Planned.
Live sandbox API	Use the VNPay sandbox for actor-eligible orders. Validate callbacks and order/amount; prevent duplicate settlement. Keep mock payments explicitly labelled. **Owner:** 524H0195. **Evidence:** Part 3 redacted provider transaction IDs and callback logs for success, cancellation/failure, invalid signature and repeated callback; automated TC-P-05–09. Credentials are excluded. Mock-only evidence does not meet this criterion. **Trace:** FR-P1/P3/P4; FR-S5. Status: Planned.
Interactive Admin dashboard	Provide R1 Revenue and R2 Hub Utilization with charts and tables. Revenue date-range and hub-selection controls refresh their respective views from authorised server data. **Owner:** 524H0191 / 524H0179. **Evidence:** Part 2 dashboard prototype; Part 3 recorded filter interaction, chart/table reconciliation against seeded data, and denied non-Admin access. Automated TC-S-03a–c. **Trace:** FR-R1/R2; FR-S2/S3. Status: Planned.

Tier criterion	Planned delivery, owner and acceptance evidence
Automated tests	Run unit tests for pricing/state rules and integration/E2E tests for orders, sandbox callbacks and the Admin dashboard. Cover at least 20 EP/BVA cases and five forbidden transitions. **Owner:** 524H0195 / 524H0191 / 524H0179. **Evidence:** Part 3 versioned commands and fixtures, commit identifier and execution reports with counts and PASS/FAIL outcomes. Planned test IDs and screenshots alone do not establish execution. **Trace:** RTM; TC-P-05–09; TC-S-03a–c. Status: Planned.

.NET 8 is retained as the existing project baseline. Microsoft lists end of support on 10 Nov 2026, before later submission dates. The team must record this lifecycle risk and obtain agreement on any runtime update; this proposal does not silently change the stack.

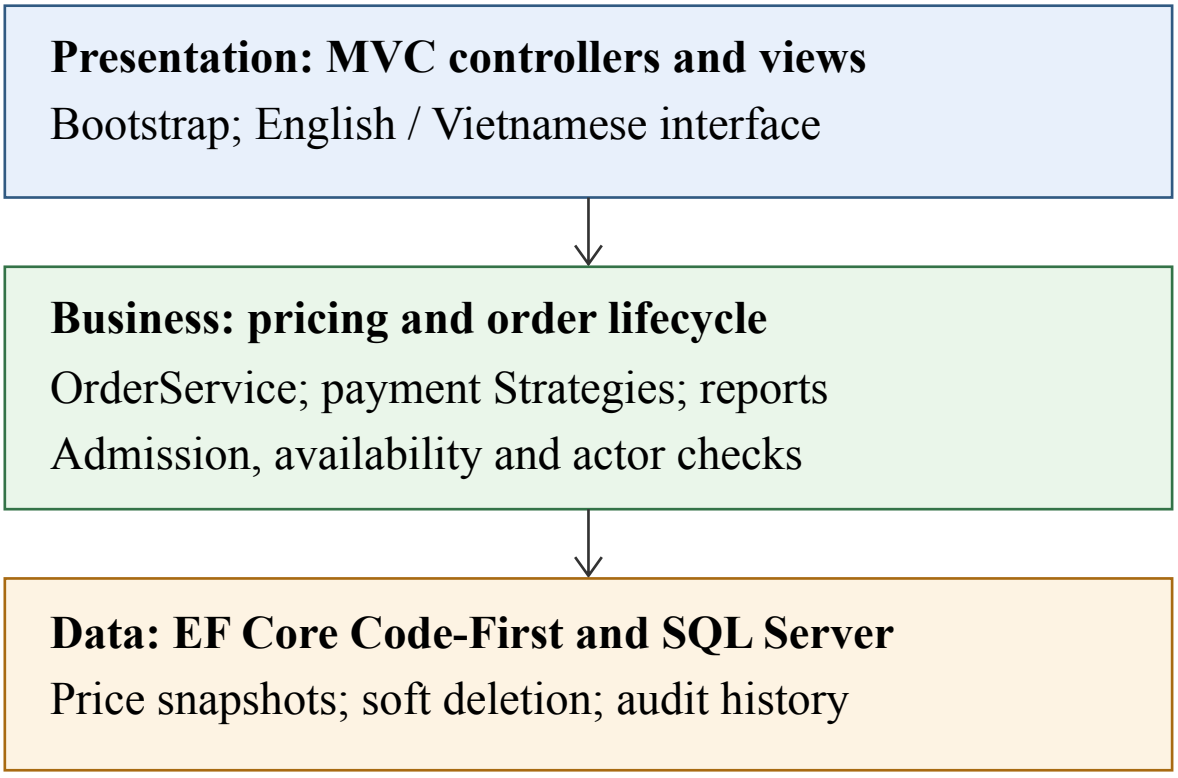


Figure 2.4.1: Proposed layers and their responsibilities

Pricing is a pure calculation over validated input and the placement snapshot. Availability, hub membership and authorisation checks belong to business services before pricing; BR12 does not require database lookups inside the arithmetic function. Payment Strategies are selected by actor type. A state transition and its audit record should be committed together.

2.5 Risks and Definition of Done

- Sandbox credentials and external availability: arrange access early; keep a clearly labelled simulated fallback, but claim Tier 4 only with real sandbox evidence.
- Change request: assess its scope in Week 5. The one-class/one-registration objective applies to a new payment method, not every possible change.
- Each requirement keeps a stable ID, source, owner and acceptance test. Design and test evidence must be linked before a requirement can receive a PASS verdict.
- Plan at least 20 EP/BVA cases covering numeric thresholds in BR1–BR12 and BR14, plus at least five forbidden-transition tests; final counts and outcomes must come from execution.

CHAPTER 3. FUNCTIONAL REQUIREMENTS

All amounts are VND. Time windows use Vietnam local time (UTC+7); start is inclusive and end exclusive. Each requirement below cites its source. R1/R2 and NFR sources are cited directly where no separate BR defines the capability. Group letters identify requirement families; the RTM gives the accountable use case and owner.

3.1 Manage Menu and Brands

FR-A1 The system shall let an administrator create a brand (name, description, operating hubs) and create its menu items (name, category, positive VND price, brand and availability).
Source: BR12; UC A; owner 524H0191.

FR-A2 The system shall let any visitor view brands and menu items filtered by hub and by category.
Source: BR12; UC A; owner 524H0191.

FR-A3 The system shall let an administrator edit brand details and a menu item’s name, category, price and brand.
Source: BR12; UC A; owner 524H0191.

FR-A4 The system shall soft-delete brands and menu items, so that historical orders and reports keep their names and prices.
Source: BR12; UC A; owner 524H0191.

FR-A5 The system shall let an administrator set item availability; unavailable or inactive items shall not be orderable.
Source: BR12; UC A; owner 524H0191.

FR-A6 The system shall reject any order line whose item belongs to a brand the chosen hub does not operate.
Source: BR12; UC A; owner 524H0191.

3.2 Manage Hubs and Capacity

FR-B1 The system shall let an administrator create a hub with a district, an address and a whole-number kitchen capacity greater than zero.
Source: BR8; UC B; owner 524H0179.

FR-B2 The system shall display hub load as 100 times the count of Confirmed or Preparing orders divided by positive integer kitchen capacity; the new order is not counted.
Source: BR8; UC B; owner 524H0179.

FR-B3 The system shall let an administrator edit a hub’s district, address and positive integer capacity, and recompute displayed load after a capacity change.
Source: BR8; UC B; owner 524H0179.

FR-B4 The system shall soft-delete a hub, removing it from ordering while preserving its historical orders and report rows.
Source: BR8; UC B; owner 524H0179.

FR-B5 The system shall refuse a new order when the chosen hub’s load is 90 % or above and shall ask the customer to choose another hub.
Source: BR8; UC B; owner 524H0179.

3.3 Manage Vouchers

FR-C1 The system shall let an administrator create a voucher with a code, a type (PERCENT or FIXED), a value, a maximum discount, a minimum subtotal, validity dates and an active flag.
Source: BR9; UC C; owner 524H0195.

FR-C2 The system shall let an administrator view, edit and deactivate vouchers.
Source: BR9; UC C; owner 524H0195.

FR-C3 The system shall accept at most one active voucher within its validity dates and check its minimum-subtotal condition against S before discounts.
Source: BR9; UC C; owner 524H0195.

3.4 Place and Manage Orders

FR-D1 The system shall let a customer build an order from the menus of one hub, with 1 to 30 lines and a quantity of 1 to 20 per line.
Source: BR12; UC D; owner 524H0195.

FR-D2 The system shall reject an order whose item subtotal is below 50,000.
Source: BR1; UC D; owner 524H0195.

FR-D3 The system shall reject an order whose delivery distance exceeds 8.0 km.
Source: BR2; UC D; owner 524H0195.

FR-D4 The system shall charge D = 15,000 for d ≤ 2 km; 15,000 + 5,000 × ceil(d − 2) for 2 < d ≤ 5; and 30,000 + 7,000 × ceil(d − 5) for 5 < d ≤ 8.
Source: BR3; UC D; owner 524H0195.

FR-D5 The system shall charge R = 20% of non-waived D when placement time in Vietnam (UTC+7) is in [11:00, 13:00) or [17:30, 19:30); otherwise R = 0.
Source: BR4; UC D; owner 524H0195.

FR-D6 The system shall add a flat 10,000 rain surcharge while the administrator’s heavy-rain flag is on, using the flag value snapshotted on the order.
Source: BR5; UC D; owner 524H0195.

FR-D7 The system shall waive the delivery fee and the rush surcharge when the item subtotal is 300,000 or more, while the rain surcharge still applies.
Source: BR6; UC D; owner 524H0195.

FR-D8 The system shall apply a Quiet Kitchen discount of 10 % of the item subtotal when the hub load at placement is below 30 %.
Source: BR7; UC D; owner 524H0195.

FR-D9 The system shall calculate PERCENT voucher V on S − Q, limited by the voucher cap and S − Q; FIXED V shall not exceed S − Q.
Source: BR9; UC D; owner 524H0195.

FR-D10 The system shall calculate S, Q, V, F = S − Q − V, D, R, R2 and raw total = F + D + R + R2 in BR10 order, retain fractional values without intermediate rounding or truncation, then round once to the nearest 1,000 VND, half up.
Source: BR10, BR11; UC D; owner 524H0195.

FR-D11 The system shall show an itemised price breakdown — subtotal, Quiet Kitchen discount, voucher, food total, delivery fee, rush surcharge, rain surcharge, rounded total — before payment.
Source: BR10; UC D; owner 524H0195.

FR-D12 The system shall allow items to be edited only while the order is Placed, and any recalculation shall reuse the stored placement snapshot.
Source: BR17; UC D; owner 524H0195.

FR-D13 The system shall allow Customer cancellation in Placed or Confirmed, Kitchen Staff rejection in Placed with a reason, and Admin cancellation in Confirmed; online-paid cancellations shall be refunded in full (simulated).
Source: BR16; UC D; owner 524H0195.

FR-D14 The system shall store a price snapshot of timestamp, hub load and rain flag at placement so that the price stays reproducible.
Source: BR17; UC D; owner 524H0195.

3.5 Manage User Accounts

FR-E1 The system shall register a Local Customer with a Vietnamese mobile number of exactly ten digits starting 03, 05, 07, 08 or 09, and a simulated OTP.
Source: BR18; UC E; owner 524H0191.

FR-E2 The system shall register a Foreign Tourist with a passport number of 6 to 9 letters or digits, a nationality, a valid email and a simulated email code.
Source: BR18; UC E; owner 524H0191.

FR-E3 The system shall enforce a unique identifier and a password of at least eight characters stored as a hash.
Source: BR18; UC E; owner 524H0191.

FR-E4 The system shall store both customer types in one schema using a discriminator, and let an administrator view, edit and deactivate accounts.
Source: BR18; UC E; owner 524H0191.

3.6 Manage Deliveries

FR-F1 The system shall let an administrator assign an available shipper to a Ready order and record the shipper and assignment time (assumption A1).
Source: BR15; UC F; owner 524H0179.

FR-F2 The system shall enforce all eight transitions and actor/condition constraints in Specification section 2C in the business layer and reject every other transition; Delivered, Cancelled and Failed are terminal.
Source: BR15; UC F; owner 524H0179.

FR-F3 The system shall require a shipper to confirm cash collected before a COD order may become Delivered.
Source: BR14; UC F; owner 524H0179.

FR-F4 The system shall let the assigned shipper move Out for Delivery to terminal Failed when the customer is unreachable, recording a reason; financial settlement follows assumption A2.
Source: BR15, BR16; UC F; owner 524H0179.

3.7 Payments

FR-P1 The system shall offer only the payment methods that belong to the customer’s actor group: MoMo, VNPay and cash for Local Customers; Apple Pay, PayPal and cash for Foreign Tourists.
Source: BR13; UC D; owner 524H0195.

FR-P2 The system shall refuse Cash on Delivery when the rounded total exceeds 1,000,000.
Source: BR14; UC D; owner 524H0195.

FR-P3 The system shall record each payment attempt with order, selected method, timestamp and success/failure outcome; only an actor-permitted Strategy may be selected.
Source: BR13; UC D; owner 524H0195.

FR-P4 The system shall integrate the live VNPay sandbox API for eligible Local Customer orders, authenticate callbacks and validate the order and amount before recording success, and process repeated callbacks idempotently.
Source: Tier 4 instructor table; BR13; UC D; owner 524H0195.

3.8 Mandatory Reports

FR-R1 The system shall produce a Revenue report for a chosen date range showing total revenue by brand category and by payment method, excluding Cancelled and Failed orders.
Source: R1; UC A; owner 524H0191.

FR-R2 The system shall produce a Hub Utilization report showing, per hub, orders handled, average preparation time from Confirmed to Ready, current load percentage and the number of Quiet Kitchen discounts granted.
Source: R2; UC B; owner 524H0179.

3.9 Cross-cutting Requirements

FR-S1 The system shall let any user switch the interface between English and Vietnamese, and the choice shall persist for the session.
Source: NFR3; UC E; owner 524H0191.

FR-S2 The system shall restrict each role to its own functions on the server, not only in the interface.
Source: NFR4; UC E; owner 524H0191.

FR-S3 The system shall provide an interactive administrator dashboard for R1 and R2 with revenue date-range and hub-selection controls that refresh the respective charts and tables from the same authorised report data.
Source: Tier 4 instructor table; R1, R2; UC A/B; owner 524H0191 / 524H0179.

FR-S4 The system shall let an administrator toggle the heavy-rain flag that the rain surcharge depends on.
Source: BR5; UC D; owner 524H0195.

FR-S5 The system shall append an audit record for each status change (actor, timestamp, old/new state) and payment attempt (actor, timestamp, method, outcome), without an application-level edit/delete operation.
Source: BR15, NFR5; UC D; owner 524H0195.

3.10 Order Lifecycle Acceptance Baseline

From → To	Allowed actor and condition
Placed → Confirmed	Kitchen Staff at the order's hub.
Placed → Cancelled	Customer; or Kitchen Staff at the hub with a rejection reason.
Confirmed → Preparing	Kitchen Staff at the order's hub.
Confirmed → Cancelled	Customer or Admin.
Preparing → Ready	Kitchen Staff at the order's hub.
Ready → Out for Delivery	Assigned Shipper; a shipper must already be assigned.
Out for Delivery → Delivered	Assigned Shipper; COD cash must be confirmed first.
Out for Delivery → Failed	Assigned Shipper; customer unreachable.

All other transitions are forbidden, including transitions from Delivered, Cancelled and Failed. Staff and shippers must be authorised for the specific order.
Source: Specification section 2C, BR14–BR16.

3.11 Pricing Precision and Boundary Checks

The nine BR10 steps are: subtotal S; Quiet Kitchen Q; voucher V; food F; delivery D and waiver; rush R; rain R2; raw sum; final half-up rounding. Preserve fractional discounts until the final rounding. With S=200,000, load=20%, d=3.2 km, time=15:00, rain off and no voucher, the specification gives Q=20,000, D=25,000 and rounded total=205,000.

Precision acceptance case: S=61,111, load=20%, a valid 10% voucher without a binding cap, d=1 km, outside rush and no rain. Q=6,111.1; V=5,499.99; raw=64,499.91; rounded total=64,000. Truncating Q/V would incorrectly produce 65,000. This is an independently calculated acceptance example, not an application execution result.

Boundary tests must include 49,999/50,000; 2/2.01, 5/5.01 and 8/8.01 km; the inclusive/exclusive rush endpoints; 299,999/300,000; loads just below/at 30% and 90%; voucher dates/minimum/cap; line and quantity limits; half-thousand totals; and COD 1,000,000/1,001,000.

CHAPTER 4. QUALITY ATTRIBUTES

NFR1–NFR4 refine the mandatory Case Study baselines; NFR5 and NFR6 are team-defined. These are measurable acceptance obligations. Their status is Planned; no application benchmark or role test is certified by this Part 1 report.

4.1 NFR1 Performance

Price calculation for a valid order of 30 lines completes in under 1.5 seconds server-side.

Verification: Time the engine with 30 valid lines on the documented test machine; record environment, elapsed times and pass/fail in Part 3.

Source and ownership: Case Study section 4B; BR12; UC D; 524H0195.

4.2 NFR2 Maintainability

Use MVC separation. Adding a payment method requires one new Strategy class and one registration, with zero edits to existing payment classes.

Verification: Add a sample method in a disposable branch; inspect the diff and rerun existing payment tests. This criterion applies to a payment-extension change request.

Source and ownership: Case Study section 4B; BR13; UC D; 524H0195.

4.3 NFR3 Usability

All required screens support English/Vietnamese and operate without horizontal page scrolling at 390, 768 and 1440 px; culture switching preserves the cart and session.

Verification: Check each screen and language at the three widths, verify resource keys and repeat switching with a non-empty cart.

Source and ownership: Case Study section 4B; UC E; 524H0191.

4.4 NFR4 Security

Passwords are hashed, never stored as plaintext; every role is refused every server action outside its authorised role and resource scope.

Verification: Inspect stored password representation; run an access matrix for both customer types, Kitchen Staff, Shipper and Admin, including other users' orders and hubs.

Source and ownership: Case Study section 4B; BR18; UC E; 524H0191.

4.5 NFR5 Auditability

100% of accepted status changes and payment attempts append a record with actor and timestamp; status changes include old/new state, payment attempts include method/outcome. No application update/delete endpoint is provided.

Verification: Reconcile accepted transitions and payment attempts against audit rows and verify ordinary roles cannot change recorded history.

Source and ownership: Team-defined; BR15; R2; UC D/F; 524H0195 / 524H0179.

4.6 NFR6 Reliability

Zero forbidden transitions are accepted; repeated pricing with identical lines and the stored placement snapshot returns identical components and rounded total.

Verification: Enumerate the state/actor matrix, include at least five forbidden transitions, and repeat boundary pricing inputs against a fixed snapshot.

Source and ownership: Team-defined; BR10, BR11, BR15, BR17; UC D/F; 524H0195 / 524H0179.

4.7 Design Implications

A pure pricing component supports deterministic tests and isolates I/O from the calculation. MVC separates interface, business services and persistence. Payment Strategies meet the extension constraint. Server-side role/resource checks support NFR4; append-only history and recorded price snapshots support NFR5/NFR6 and the preparation-time calculation in R2.

CHAPTER 5. AMBIGUITY LOG

The following are working assumptions, not instructor approvals. The Specification is authoritative for fixed rules; an ambiguity cannot override a stated threshold, calculation order or transition. The same A IDs are used in the workbook and pitch.

5.1 A1 Shipper assignment

Gap: Section 2C leaves the assignment mechanism open.

Assumption: An Admin manually selects an available shipper for a Ready order. The system stores assignment time and shipper. Automatic dispatch is outside the initial scope.

Impact: FR-F1; UC F; assignment screen and service.

Instructor answer/status: Not asked; working assumption pending instructor review.

5.2 A2 Settlement after Failed delivery

Gap: Section 2C leaves the financial result of Failed delivery open.

Assumption: Refund online-paid orders in full in the simulated ledger; collect no COD cash. Preserve Failed as its own terminal state and exclude it from R1 revenue.

Impact: FR-F4; UC F/D; payment/refund policy and R1.

Instructor answer/status: Not asked; working assumption pending instructor review.

5.3 A3 Tourist delivery address

Gap: Hotel and room are allowed, but a field structure is not prescribed.

Assumption: Store street/address text plus optional hotel name and room. Require a usable delivery address for either customer type; tourists are not required to supply a Vietnamese phone.

Impact: FR-E2 and FR-D1; address validation and checkout.

Instructor answer/status: Not asked; working assumption pending instructor review.

5.4 A4 Invalid voucher user flow

Gap: BR9 specifies eligibility but not whether a refused voucher should block the whole order.

Assumption: Show the refusal and the corrected breakdown with V=0; require the customer to review and confirm that total before placing the order. Never silently submit a higher total.

Impact: FR-C3 and FR-D9/D11; checkout confirmation.

Instructor answer/status: Not asked; working assumption pending instructor review.

5.5 A5 Catalogue price change before a Placed edit

Gap: BR17 fixes timestamp/load/rain but does not explicitly say which item prices apply when the catalogue changes before an edit.

Assumption: Existing lines retain their recorded unit prices; newly added items use the current catalogue price after availability/membership validation. Recalculation still uses the original timestamp, load and rain snapshot.

Impact: FR-D12/D14 and FR-A4; order-line price storage and edit tests.

Instructor answer/status: Not asked; working assumption pending instructor review.

Changes to these assumptions require updating affected FRs, design mappings and acceptance tests. The current portal screenshot names SaigonBite consistently; no product-name conflict is asserted. BR3’s distance discontinuity and BR11’s final-only rounding are fixed rules, not open assumptions.

CHAPTER 6. REQUIREMENTS TRACEABILITY MATRIX

6.1 Purpose and Status

The seed matrix contains 47 FRs and 6 NFRs. Every row has an accountable use case, planned implementation mapping and acceptance test reference. Component names are proposed mappings, to be reconciled with Part 2 design and Part 3 code. All verification verdicts remain Planned; tests listed here have not been executed as part of Part 1.

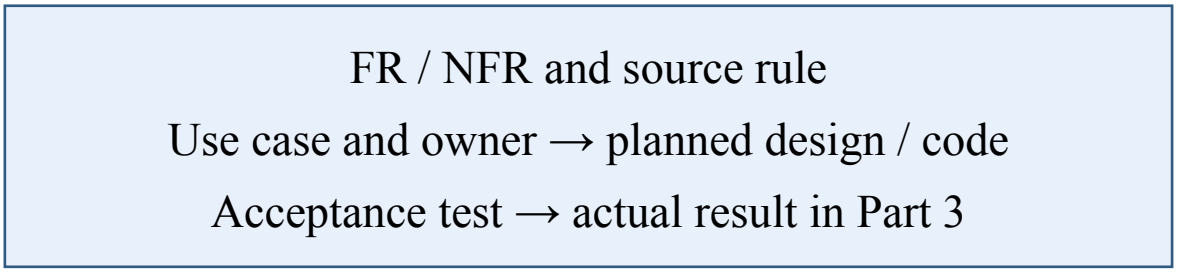


Figure 6.1.1: Traceability from source requirement to later verification

6.2 Functional Requirement Mappings

Requirement	Planned trace and acceptance test
FR-A1	UC A; owner 524H0191. Source: BR12. Design: Brand entity, menu admin; code: BrandsController. Tests: TC-A-01. Verdict: Planned.
FR-A2	UC A; owner 524H0191. Source: BR12. Design: Catalogue view; code: BrandsController. Tests: TC-A-02. Verdict: Planned.
FR-A3	UC A; owner 524H0191. Source: BR12. Design: Menu edit form; code: BrandsController. Tests: TC-A-03. Verdict: Planned.
FR-A4	UC A; owner 524H0191. Source: BR12. Design: Soft-delete filter; code: DbContext filters. Tests: TC-A-04. Verdict: Planned.
FR-A5	UC A; owner 524H0191. Source: BR12. Design: Availability flag; code: MenuItemService. Tests: TC-A-05. Verdict: Planned.

Requirement	Planned trace and acceptance test
FR-A6	UC A; owner 524H0191. Source: BR12. Design: Hub-brand relation; code: OrderService. Tests: TC-A-06. Verdict: Planned.
FR-B1	UC B; owner 524H0179. Source: BR8. Design: Hub entity; code: HubsController. Tests: TC-B-01. Verdict: Planned.
FR-B2	UC B; owner 524H0179. Source: BR8. Design: Derived load; code: HubLoadCalculator. Tests: TC-B-02. Verdict: Planned.
FR-B3	UC B; owner 524H0179. Source: BR8. Design: Capacity edit; code: HubsController. Tests: TC-B-03. Verdict: Planned.
FR-B4	UC B; owner 524H0179. Source: BR8. Design: Soft-delete filter; code: DbContext filters. Tests: TC-B-04. Verdict: Planned.
FR-B5	UC B; owner 524H0179. Source: BR8. Design: Admission rule; code: OrderAdmissionService. Tests: TC-B-05. Verdict: Planned.
FR-C1	UC C; owner 524H0195. Source: BR9. Design: Voucher entity; code: VouchersController. Tests: TC-C-01. Verdict: Planned.
FR-C2	UC C; owner 524H0195. Source: BR9. Design: Voucher admin; code: VouchersController. Tests: TC-C-02. Verdict: Planned.
FR-C3	UC C; owner 524H0195. Source: BR9. Design: Voucher validation; code: VoucherService. Tests: TC-C-03. Verdict: Planned.
FR-D1	UC D; owner 524H0195. Source: BR12. Design: Cart, line limits; code: OrdersController. Tests: TC-D-01. Verdict: Planned.
FR-D2	UC D; owner 524H0195. Source: BR1. Design: Minimum order; code: PricingEngine. Tests: TC-D-02, 03. Verdict: Planned.

Requirement	Planned trace and acceptance test
FR-D3	UC D; owner 524H0195. Source: BR2. Design: Coverage check; code: PricingEngine. Tests: TC-D-04, 05. Verdict: Planned.
FR-D4	UC D; owner 524H0195. Source: BR3. Design: Distance tiers; code: PricingEngine. Tests: TC-D-06…08. Verdict: Planned.
FR-D5	UC D; owner 524H0195. Source: BR4. Design: Rush windows; code: PricingEngine. Tests: TC-D-09…11. Verdict: Planned.
FR-D6	UC D; owner 524H0195. Source: BR5. Design: Rain surcharge; code: PricingEngine. Tests: TC-D-12, 13. Verdict: Planned.
FR-D7	UC D; owner 524H0195. Source: BR6. Design: Free-delivery waiver; code: PricingEngine. Tests: TC-D-14, 15. Verdict: Planned.
FR-D8	UC D; owner 524H0195. Source: BR7. Design: Quiet Kitchen rule; code: PricingEngine. Tests: TC-D-16, 17. Verdict: Planned.
FR-D9	UC D; owner 524H0195. Source: BR9. Design: Voucher application; code: PricingEngine. Tests: TC-D-18…20. Verdict: Planned.
FR-D10	UC D; owner 524H0195. Source: BR10, BR11. Design: Nine-step order; code: PricingEngine. Tests: TC-D-21, 22. Verdict: Planned.
FR-D11	UC D; owner 524H0195. Source: BR10. Design: Breakdown view; code: OrdersController. Tests: TC-D-23. Verdict: Planned.
FR-D12	UC D; owner 524H0195. Source: BR17. Design: Edit while Placed; code: OrderService. Tests: TC-D-24. Verdict: Planned.
FR-D13	UC D; owner 524H0195. Source: BR16. Design: Cancellation rules; code: OrderStateMachine. Tests: TC-D-25, 26. Verdict: Planned.

Requirement	Planned trace and acceptance test
FR-D14	UC D; owner 524H0195. Source: BR17. Design: Price snapshot; code: Order entity. Tests: TC-D-27. Verdict: Planned.
FR-E1	UC E; owner 524H0191. Source: BR18. Design: Local validation; code: AccountController. Tests: TC-E-01, 02. Verdict: Planned.
FR-E2	UC E; owner 524H0191. Source: BR18. Design: Tourist validation; code: AccountController. Tests: TC-E-03, 04. Verdict: Planned.
FR-E3	UC E; owner 524H0191. Source: BR18. Design: Hash + uniqueness; code: AccountService. Tests: TC-E-05. Verdict: Planned.
FR-E4	UC E; owner 524H0191. Source: BR18. Design: Discriminator schema; code: DbContext. Tests: TC-E-06. Verdict: Planned.
FR-F1	UC F; owner 524H0179. Source: BR15. Design: Shipper assignment; code: DeliveriesController. Tests: TC-F-01. Verdict: Planned.
FR-F2	UC F; owner 524H0179. Source: BR15. Design: State machine; code: OrderStateMachine. Tests: TC-F-02, 03. Verdict: Planned.
FR-F3	UC F; owner 524H0179. Source: BR14. Design: Cash confirmation; code: DeliveryService. Tests: TC-F-04. Verdict: Planned.
FR-F4	UC F; owner 524H0179. Source: BR15, BR16. Design: Failure path; code: DeliveryService. Tests: TC-F-05. Verdict: Planned.
FR-P1	UC D; owner 524H0195. Source: BR13. Design: Actor-method map; code: PaymentStrategyResolver. Tests: TC-P-01, 02. Verdict: Planned.
FR-P2	UC D; owner 524H0195. Source: BR14. Design: COD ceiling; code: CodStrategy. Tests: TC-P-03. Verdict: Planned.

Requirement	Planned trace and acceptance test
FR-P3	UC D; owner 524H0195. Source: BR13. Design: Payment record; code: PaymentService. Tests: TC-P-04. Verdict: Planned.
FR-P4	UC D; owner 524H0195. Source: Tier 4 instructor table; BR13. Design: Sandbox adapter and callback validation; code: VnPayStrategy / PaymentService. Tests: TC-P-05–09. Verdict: Planned.
FR-R1	UC A; owner 524H0191. Source: R1. Design: Revenue aggregation; code: ReportService. Tests: TC-RT-01, 02. Verdict: Planned.
FR-R2	UC B; owner 524H0179. Source: R2. Design: Utilisation aggregation; code: ReportService. Tests: TC-RT-03, 04. Verdict: Planned.
FR-S1	UC E; owner 524H0191. Source: NFR3. Design: Resource files; code: Localisation. Tests: TC-S-01. Verdict: Planned.
FR-S2	UC E; owner 524H0191. Source: NFR4. Design: Authorisation policy; code: Policies. Tests: TC-S-02. Verdict: Planned.
FR-S3	UC A/B; owner 524H0191 / 524H0179. Source: Tier 4 instructor table; R1, R2. Design: Dashboard; code: AdminController. Tests: TC-S-03a–c. Verdict: Planned.
FR-S4	UC D; owner 524H0195. Source: BR5. Design: Rain flag; code: AdminController. Tests: TC-S-04. Verdict: Planned.
FR-S5	UC D; owner 524H0195. Source: BR15, NFR5. Design: Audit writer; code: OrderStatusHistory. Tests: TC-S-05. Verdict: Planned.

6.3 Quality Attribute Mappings

Requirement	Planned trace and acceptance test
NFR1	UC D; owner 524H0195. Source: Case Study section 4B; BR12. Design/code: pure PricingEngine. Test: TC-NFR-1. Time the engine with 30 valid lines on the documented test machine; record environment, elapsed times and pass/fail in Part 3. Verdict: Planned.
NFR2	UC D; owner 524H0195. Source: Case Study section 4B; BR13. Design/code: IPaymentStrategy and registration. Test: TC-NFR-2. Add a sample method in a disposable branch; inspect the diff and rerun existing payment tests. This criterion applies to a payment-extension change request. Verdict: Planned.
NFR3	UC E; owner 524H0191. Source: Case Study section 4B. Design/code: responsive views and localisation. Test: TC-NFR-3. Check each screen and language at the three widths, verify resource keys and repeat switching with a non-empty cart. Verdict: Planned.
NFR4	UC E; owner 524H0191. Source: Case Study section 4B; BR18. Design/code: identity and authorisation policies. Test: TC-NFR-4. Inspect stored password representation; run an access matrix for both customer types, Kitchen Staff, Shipper and Admin, including other users' orders and hubs. Verdict: Planned.
NFR5	UC D/F; owner 524H0195 / 524H0179. Source: Team-defined; BR15; R2. Design/code: audit/history writer. Test: TC-NFR-5. Reconcile accepted transitions and payment attempts against audit rows and verify ordinary roles cannot change recorded history. Verdict: Planned.
NFR6	UC D/F; owner 524H0195 / 524H0179. Source: Team-defined; BR10, BR11, BR15, BR17. Design/code: state machine and pricing snapshot. Test: TC-NFR-6. Enumerate the state/actor matrix, include at least five forbidden transitions, and repeat boundary pricing inputs against a fixed snapshot. Verdict: Planned.

6.4 Change Control

The report and workbook are generated from config/part1.json. Every new or changed ID must appear in both and keep its source/owner/test mapping. Part 2 adds approved diagrams; Part 3 records actual components, execution evidence and PASS/FAIL verdicts. A test identifier is a plan until its execution record exists.

APPENDIX A. GENERATIVE AI USAGE LOG

AI assistance is disclosed below using the columns required by Specification section 7. Historical DSH activity is summarised from the existing project log; its original full prompts and outputs have not been located in this workspace. The team must recover the original records for the instructor's 48-hour request requirement. A reconstructed summary must not be represented as an original transcript.

A.1 DeepSeek Harness (DSH), 23 Oct 2026

Required field	Disclosure
Tool and date	DeepSeek Harness (DSH), 23 Oct 2026
Purpose	Earlier draft and project scaffolding
Prompt (summary)	Earlier project log describes extraction of BRs, FR drafting, architecture and rendering. This is a summary from that log, not a recovered raw prompt.
What was kept or changed	Retained the useful document structure and stable FR IDs; corrected conflicting thresholds, ownership and unsupported claims during this revision.
How verified	Compared current requirements with the three course PDFs; earlier raw prompts/outputs are not available in this workspace.

A.2 OpenAI Codex, 24 Oct 2026

Required field	Disclosure
Tool and date	OpenAI Codex, 24 Oct 2026
Purpose	Strict Part 1 audit
Prompt (summary)	User requested a lecturer-style audit of existing Part 1 against the screenshot, course requirements, templates and TDTU skills.
What was kept or changed	Recorded faults and repair priorities in PART1_AUDIT_2026-10-24.md; did not use the provisional score as an official course mark.
How verified	Read the course PDFs, extracted workbook cells, rendered report/slides, checked links and calculated the rounding counterexample independently.

A.3 OpenAI Codex, 24 Oct 2026

Required field	Disclosure
Tool and date	OpenAI Codex, 24 Oct 2026
Purpose	Correct and synchronise Part 1 deliverables
Prompt (summary)	User: “ok giờ bạn cứ sửa sao cho chuẩn nhất đi đẻ hoàn thiện part 1”.
What was kept or changed	One shared FR/NFR/ambiguity register, corrected report/workbook/pitch and explicit planned verification. No invented signatures or instructor answers.
How verified	Build and source-consistency checks; rendered output checks are recorded in the Part 1 verification note. Live pitch, sandbox and application tests are outside this document revision.

The available Codex request excerpts and work summary are retained in docs/appendix/prompts/CODEX_2026-10-24.md; that file is labelled as excerpts and a summary. No full historical transcript or unrecorded human review is claimed.

APPENDIX B. CONTRIBUTION STATEMENT

Planned responsibility is set out below. The course requires a statement signed by all members at the start of Part 3. Signatures, dates and actual contribution percentages must be supplied by the members; none are generated here.

Member	Planned responsibility
Ngô Lâm Tiến 524H0195	Main UC D: Place and Manage Orders. Secondary: C - Manage Vouchers; Pricing Engine (BR1-BR12); Payment Strategy Module. Reports: none.
Nguyễn Công Thành 524H0191	Main UC A: Manage Menu & Brands. Secondary: E - Manage User Accounts. Reports: R1 - Revenue Report.
Nguyễn Trọng Phú 524H0179	Main UC B: Manage Hubs & Capacity. Secondary: F - Manage Deliveries; Change Request Impact Note. Reports: R2 - Hub Utilization Report.

Ngô Lâm Tiến (524H0195)

Signature: _____ Date: _____

Nguyễn Công Thành (524H0191)

Signature: _____ Date: _____

Nguyễn Trọng Phú (524H0179)

Signature: _____ Date: _____

REFERENCES

Ton Duc Thang University, Faculty of Information Technology (2026). *SE Final Project Specification: SaigonBite*, version 1.0, Sem01_2627. BR1–BR18, Part 1 checklist and rubric.

Ton Duc Thang University (2026). *SaigonBite Case Study: Cloud-Kitchen Food Delivery Hub*, Sem01_2627. Context, FR1–FR9 and quality baselines.

Ton Duc Thang University (2026). *SaigonBite Student Survival Guide*, Sem01_2627. Milestones and presentation guidance.

Course instructor. *Technology tiers (from the specification; the instructor may edit the blue caps)*. Image supplied by the student on 24 Oct 2026; publication date not shown. Archived at docs/references/instructor-technology-tiers.png. Current ceilings: 7.0, 7.5, 8.5, 9.5 and 10.0.

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